

PPI-hotspot^{ID}: A Method for Detecting Protein-Protein Interaction Hot Spots from the Free Protein Structure

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Abstract

Experimental detection of residues critical for protein-protein interactions (PPI) is a timeconsuming, costly, and labor-intensive process. Hence, high-throughput PPI-hot spot prediction methods have been developed, but they have been validated using relatively small datasets, which may compromise their predictive reliability. Here, we introduce PPI-hotspot^{ID}, a novel method for identifying PPI-hot spots using the free protein structure, and validated it on the largest collection of experimentally confirmed PPI-hot spots to date. We explored the possibility of detecting PPI-hot spots using (i) FTMap in the PPI mode, which identifies hot spots on protein-protein interfaces from the *free* protein structure, and (ii) the interface residues predicted by AlphaFold-Multimer. PPI-hotspot^{ID} yielded better performance than FTMap and SPOTONE, a webserver for predicting PPI-hotspots given the protein sequence. When combined with the AlphaFold-Multimer-predicted interface residues, PPI-Hotspot^{ID}, also yielded better performance than either method alone. Furthermore, we experimentally verified several PPI-hot spots of eukaryotic elongation factor 2 predicted by PPI-hotspot^{ID}. Notably, PPI-hotspot^{ID} unveils PPI-hot spots that are not obvious from complex structures, which only reveal interface residues, thus overlooking PPI-hot spots in *indirect* contact with binding partners. Thus, PPI-hotspot^{ID} serves as a valuable tool for understanding the mechanisms of PPIs and facilitating the design of novel drugs targeting these interactions. A freely accessible web server is available at <https://ppihotspotid.limlab.dnsalias.org/> and the source code for PPI-hotspot^{ID} at <https://github.com/wrigjz/ppihotspotid/>.

eLife assessment

The manuscript presents a machine-learning method to predict protein hotspot residues. The validation is **incomplete**, along with the misinterpretation of the results with other current methods like FTMap.

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Introduction

Protein-protein interactions (PPIs) play a crucial role in cellular physiology, and their dysregulation is associated with various diseases,^{1,2} such as cancer,² infectious diseases, and neurodegenerative diseases.³ Identifying residues critical for PPIs (termed PPI-hot spots) is important for elucidating protein function and designing targeted biomedical interventions.^{4,5} Conventionally, PPI-hot spots are defined as residues whose mutations to alanine cause ≥ 2 kcal/mol drop in the protein binding free energy.^{6–11} However, this definition, based on measuring the binding free energy change upon mutation to alanine, limits the number of experimentally determined PPI-hot spots. Hence, PPI-hot spots have been more broadly defined to include residues whose mutations, not necessarily to alanine, significantly impair/disrupt PPIs,^{12,13} as detected by experimental methods such as coimmunoprecipitation and yeast two-hybrid screening. As each mutant must be purified and analyzed separately, experimental detection of PPI-hot spots is time-consuming, costly, and labor-intensive.

To enable large-scale detection of PPI-hot spots, high-throughput PPI-hot spot prediction methods have been developed. They generally fall into 2 categories:¹⁴ (1) Methods that compute the binding energy/free energy difference between the wild-type protein and a mutant using classical force fields or empirical scoring functions.^{11,15–23} (2) Methods that employ classifiers such as nearest neighbor, support vector machines, decision trees, Bayesian/neural networks, random forest, and ensemble machine-learning models using various features including conservation, secondary structure, solvent-accessible surface area, and atom density.^{14,24–36} Most of the PPI-hot spot prediction methods rely on the protein complex structure and some are accessible via webserver; e.g., Hotpoint,³⁷ KFC2,³⁸ PredHS,³⁹ and PredHS2.⁴⁰ Fewer methods use only the *free* protein structure^{41–45} or sequence,^{34,46–54} and SPOTONE (hot SPOTs ON protein complexes with Extremely randomized trees) is available as a web server. SPOTONE⁵³ predicts PPI-hot spots from the protein sequence using residuespecific features such as atom type, amino acid (aa) properties, secondary structure propensity, and mass-associated values to train an ensemble of extremely randomized trees.

The PPI-hot spot prediction methods have mostly been trained, validated, and tested on data from the Alanine Scanning Energetics database (ASEdb)⁵⁵ and/or the Structural Kinetic and Energetic database of Mutant Protein Interactions (SKEMPI) 2.0 database.⁵⁶ However, antibody-antigen interactions have different sequence and structural characteristics compared to non-antibody PPIs.⁵⁷ Therefore, our focus is exclusively only on non-antibody proteins in this study. The ASEdb contains 96 PPI-hot spots from 26 proteins. SKEMPI 2.0, which includes single-point mutations not necessarily to alanine that decrease the protein binding free energy by ≥ 2 kcal/mol, has 343 PPI-hot spots from 117 distinct proteins, 40 of which overlap with ASEdb. Altogether, ASEdb and SKEMPI contain 399 distinct PPI-hot spots in 132 proteins. To increase this number of distinct PPI-hot spots, we have expanded the definition of PPI-hot spots to include mutations in UniProtKB⁵⁸ that have been manually curated as significantly impairing/disrupting PPIs.¹³ This expanded definition led to the creation of the PPI-Hotspot^{DB} database, which contains 4,039 experimentally determined PPI-hot spots in 1,893 proteins. To calibrate PPI-hot spot prediction methods using the free protein structure, a benchmark was derived from PPI-Hotspot^{DB}.¹³ This benchmark called PPI-Hotspot+PDB^{BM} contains nonredundant proteins with free structures and known PPI-hot spots. The proteins in PPI-Hotspot+PDB^{BM} share $< 60\%$ sequence identity, which has been shown to be a reasonable threshold for grouping domains with similar functions.¹³

Our aim is to develop a method for identifying PPI-hot spots in non-antibody proteins using free protein structures. First, we updated the PPI-Hotspot+PDB^{BM} benchmark and constructed a dataset comprising 158 nonredundant proteins with free structures harboring 414 experimentally

known PPI-hot spots and 504 PPI-nonhot spots (see **Methods**). Using this dataset, we applied an automatic machine-learning framework that automates the machinelearning pipeline to detect PPI-hot spots using the aa type as well as structural, energetic, and evolutionary features of each residue in a protein. The resulting prediction model, named PPI-Hotspot^{ID}, identifies PPI-hot spots using an ensemble of classifiers and only 4 residue features (conservation, aa type, solvent-accessible surface area (SASA), and gas-phase energy, ΔG^{gas}). We explored the possibility of detecting PPI-hot spots using the FTMap server in the PPI mode, which identifies hot spots on protein–protein interfaces from *free* protein structures.⁴⁵ These hot spots are identified by consensus sites - regions that bind multiple probe clusters.^{42, 45, 59} Such regions are deemed to be important for any interaction involving that region of the target, independent of partner protein.⁴² PPI-hot spots were identified as residues in van der Waals (vdW) contact with probe ligands within the largest consensus site containing the most probe clusters. We also explored the possibility of detecting PPI-hot spots using the interface residues predicted by AlphaFold-Multimer,⁶⁰ which has been shown to outperform current docking methods in predicting protein-protein complexes. Finally, we illustrated the utility of PPI-Hotspot^{ID} by applying it to detect PPI-hot spots of eukaryotic elongation factor 2 (eEF2), a translation factor essential for peptide elongation, and experimentally verified the predictions.

Results

Evaluating the Performance of PPI-Hot Spot Detection Methods.

The goal of PPI-hotspot^{ID} is to detect true PPI-hot spots rather than true PPI-nonhot spots in proteins. Hence, we assessed the performance of PPI-hotspot^{ID} by computing the sensitivity/recall (the fraction of true PPI-hot spots correctly identified),

$$\text{Sensitivity} = \frac{TP}{TP+FN} \quad (1)$$

the fraction of predicted PPI-hot spots that are true PPI-hot spots; i.e.,

$$\text{Precision} = \frac{TP}{TP+FP} \quad (2)$$

and the F1-score, which combines recall and precision:

$$F1 = \frac{2 \times (\text{sensitivity} \times \text{precision})}{(\text{sensitivity} + \text{precision})} = \frac{2TP}{2TP+FP+FN} \quad (3)$$

Since our dataset also contains true PPI-nonhot spots, we calculated the specificity (the fraction of true PPI-nonhot spots correctly identified):

$$\text{Specificity} = \frac{TN}{TN+FP} \quad (4)$$

In eqs 1–4, TP (true positives) or TN (true negatives) is the number of correctly predicted PPI-hot spots or PPI-nonhot spots, and FP (false positives) or FN (false negatives) is the number of wrongly predicted PPI-hot spots or PPI-nonhot spots.

Performance of PPI-hotspot^{ID} vs. FTMap and SPOTONE.

We compared the performance of PPI-Hotspot^{ID}, FTMap,⁴⁵ and SPOTONE⁵³ using a dataset containing 414 true PPI-hot spots and 504 nonhot spots (see **Methods**). Note that the 414 true PPI-hot spots represent only 2% of the total number of residues (21,722) in the 158 proteins. Given the free protein structure, PPI-Hotspot^{ID} and SPOTONE⁵³ predict PPI-hot spots based on a probability threshold (> 0.5). FTMap, in the PPI mode, detects PPI-hot spots as consensus sites/regions on the protein surface that bind multiple probe clusters.⁵⁹ Residues in vdW contact with probe molecules within the largest consensus site were compared with PPI-hotspot^{ID}/SPOTONE predictions. Residues *not* classified as PPI-hot spots by each method were considered as PPI-nonhot spots. **Table 1** summarizes the results for our dataset, with the F1 score in parentheses representing the mean validation F1 score. Compared to FTMap/SPOTONE, PPI-Hotspot^{ID} detected a much higher fraction of true positives (0.67 vs. 0.07/0.10) and achieved a significantly higher F1 score (0.71 vs. 0.13/0.17).

To elucidate the differences between the PPI-hot spots predicted by PPI-Hotspot^{ID} and those by FTMap or SPOTONE, we compared their respective true positive predictions. The Venn diagram in **Table 1** shows a substantial overlap in true positives between FTMap or SPOTONE and PPI-Hotspot^{ID}: FTMap shared 23/30 true positives with PPI-Hotspot^{ID}, whereas SPOTONE shared 34/40 with PPI-Hotspot^{ID}, but only 3 with FTMap. Only 3 true positives were predicted by all 3 methods. PPI-Hotspot^{ID} identified many true positives that were not detected by FTMap or SPOTONE probably because it employed aspects not considered by FTMap or SPOTONE such as the gas-phase energy, ΔG^{gas} (see **Discussion**). Furthermore, SPOTONE defined true negatives as residues whose mutation to alanine led to protein binding free energy changes ($\Delta\Delta G^{bind}$) of ≤ 2.0 kcal/mol, whereas we defined true negatives as residues whose alanine/nonalanine mutations resulted in negligible protein binding free energy changes ($\Delta\Delta G^{bind} < 0.5$ kcal/mol) or did not perturb PPIs in immunoprecipitation or GST pull-down assays (see **Methods**).

Interface vs. Noninterface PPI-hotspots.

We can estimate the fraction of PPI-hotspots at the protein interface for 74 of the 158 nonredundant proteins in our dataset with complex structures. These 74 proteins harboring 243 true PPI-hotspots form 78 PPI pairs. Using the UniProt codes of each protein and its binding partner, we identified all complex structures in the PDB. Based on the complex structures of each PPI pair, we classified a PPI-hotspot as an interface one if it formed hydrogen bonds or vdW contacts with the partner protein,⁶¹ otherwise it was deemed a *noninterface* PPI-hotspot. Among the 243 true PPI-hotspots, 67 (27.6%) lacked such contacts across the protein interface. For these 74 proteins, PPI-hotspot^{ID} predicted 240 PPI-hot spots, out of which 43 (18%) are *noninterface* PPI-hot spots, SPOTONE identified only 5 noninterface PPI-hot spots, whereas FTMap did not predict any. For example, the complex structure of interleukin-4 bound to interleukin-4 receptor subunit α (PDB: 1IAR)⁶² in **Figure 1** revealed 3 interface PPI-hot spots (E13, R92 and N93) and 2 noninterface ones (K127 and Y128). Based on the free structure of interleukin-4 (PDB: 1BBN),⁶³ PPI-Hotspot^{ID} identified all 5 true positives, SPOTONE detected an interfacial PPI-hot spot (N93), whereas FTMAP failed to identify any true positives.

Performance of AlphaFold-Multimer, PPI-Hotspot^{ID} and Their Combination in Predicting PPI-Hot Spots.

To assess the possibility of detecting PPI-hot spots using the interface residues predicted by AlphaFold-Multimer⁶⁰ as PPI-hot spots when complex structures are unavailable, we focused on 48 “unsolved” AB complex structures involving 47 proteins in the PPI-Hotspot+PDB^{BM(1.1)}, as one of the proteins, human neurotrophin (UniProtID P20783, PDB 1nt30A) interacted with two different proteins (UniProtID Q16288 and P17643). These 48 *unsolved* complex structures contain

Table 1.

Performance of the PPI-Hotspot^{ID} vs. FTMap and SPOTONE^a [↗](#)

Method	PPI-Hotspot ^{ID}	FTMap	SPOTONE
TP	278	30	40
FN	136	384	374
TN	417	487	481
FP	87	17	23
Sensitivity	0.67	0.07	0.10
Precision	0.76	0.64	0.64
F1	0.71 (0.66) ^b	0.13	0.17
Specificity	0.83	0.97	0.95

Venn diagram illustrating the overlap of PPI-Hotspot^{ID} (blue), FTMap (red), and SPOTONE (green). The counts are: PPI-Hotspot^{ID} only: 224; FTMap only: 7; SPOTONE only: 6; PPI-Hotspot^{ID} & FTMap: 20; PPI-Hotspot^{ID} & SPOTONE: 31; FTMap & SPOTONE: 0; All three: 3.

Table 2.

Performance of AlphaFold-Multimer, PPI-Hotspot^{ID}, and their combination for 48 “unsolved” complex structures

Method	AlphaFold2-multimer	PPI-Hotspot ^{ID}	AlphaFold2-multimer + PPI-Hotspot ^{ID}
TP	37	52	63
FN	53	38	27
TN	35	29	24
FP	10	16	21
Sensitivity	0.41	0.58	0.70
Precision	0.79	0.77	0.75
F1	0.54	0.66 ^b	0.72
Specificity	0.78	0.64	0.53

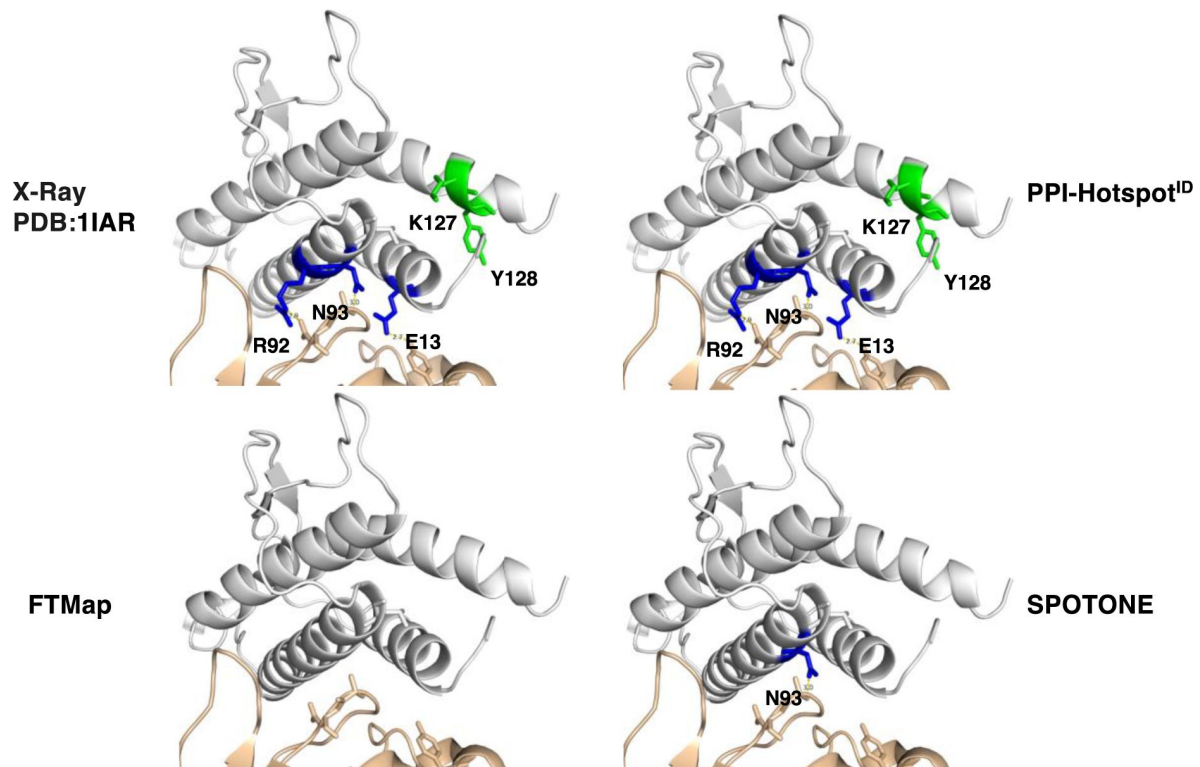


Figure 1.

(Top left) The X-ray structure (PDB: 1IAR) of interleukin-4 (**grey**) in complex with interleukin-4 receptor subunit alpha (wheat) with five PPI-hot spots; interface PPI-hot spots (E13, R92, and N93) are in blue and the non-interface ones (K127 and Y128) are in green. The PPI-hot spot numberings are based on the interleukin-4 free structure (PDB: 1BBN). The correct predictions of PPI-hotspot^{ID} (top right), FTMap (bottom left) and SPOTONE (bottom right) are mapped to the corresponding residues of the complex structure.

90 PPI-hot spots and 45 nonhot spots. We employed the protein A structure sequence from the PPI-Hotspot+PDB^{BM(1.1)} and the entire protein B sequence from UniProtKB⁵⁸ as inputs for the AlphaFold-Multimer module in ColabFold.⁶⁴ This generated model structures for each AB complex. Interface residues were defined based on the AMBER-relaxed model structure with the highest pTM score using a cutoff distance of 5 Å reflecting residues in close contact. Interface residues were predicted as PPI-hot spots and noninterface residues as nonhot spots.

In identifying PPI-hot spots using PPI-Hotspot^{ID}, we first excluded 90 true PPI-hot spots and 45 nonhot spots belonging to 47 proteins lacking complex PDB structures from our dataset. We then used an automatic machine-learning framework to train an ensemble of machine learning models using 4 features (k^C , aa residue type, SASA_i, and ΔG_i^{gas}) on the true PPI-hot spots and nonhot spots in the remaining 111 proteins in our dataset. The final ensemble model was used to identify PPI-hot spots in the 47 proteins lacking complex structures in our dataset. The resulting sensitivity (0.58) and F1 score (0.66) were lower than those in **Table 1** using the full dataset. Nevertheless, they were greater than those achieved using AlphaFold-Multimer-predicted interface residues as PPI-hot spots (0.41). When we combined the PPI-Hotspot^{ID}-predicted PPI-hot spots with the AlphaFold-Multimer-predicted interface residues, the resulting sensitivity (0.70) and F1 values (0.72) were higher than those obtained by each method alone. This indicates that PPI-Hotspot^{ID} can identify true PPI-hot spots that reside outside the protein-protein interface.

Experimental Verification of PPI-

Hotspot^{ID}'s Predictions in Human eEF2.

We experimentally verified predictions made by PPI-Hotspot^{ID} by using it to detect the PPI-hot spots of eEF2, an essential translation factor that hydrolyzes GTP to catalyze peptide elongation. Binding of cytoplasmic polyadenylation element-binding protein-2 (CPEB2) to eEF2 may interfere with conformational changes of eEF2 on ribosomes, thereby affecting the efficiency of eEF2-mediated GTP hydrolysis, and slowing down translation of hypoxiainducible factor (HIF)-1α mRNA.⁶⁵ No eEF2-CPEB2 complex structure has been solved, but a 5-Å electron microscopy structure of eEF2 (PDB 4v6x-A)⁶⁶ is available. Using the CPEB2 N-terminus for a yeast two-hybrid screen, a positive clone containing the eEF2 residues 717–803 had been identified and subsequent co-IP assay revealed a CPEB2-binding domain comprising eEF2 residues 743–817.⁶⁵ Thus, we focused on this domain, which shares ≤20% sequence identity with the 158 nonredundant proteins in our dataset, in predicting PPI-hot spots. Based on the free eEF2 structure (PDB 4v6x-A),⁶⁶ PPI-hotspot^{ID} predicted F794 as the PPI-hot spot with the highest probability of 0.67. So, we chose to test F794 and 7 other predicted PPI-hot spots (L763, R767, G768, G778, T779, R801, A808) that were > 12 Å from F794, as well as 4 predicted PPI-nonhot spots (E773, P789, V790, Q807).

To validate PPI-Hotspot^{ID}'s predictions, we mutated the aforementioned predicted PPI-hot spots and PPI-nonhot spots in mouse eEF2 (meEF2), which shares 99% sequence identity with human eEF2. The generated eEF2 mutants (L763A, ⁷⁶⁶AAA⁷⁶⁸, E773Q, ⁷⁷⁸AAA⁷⁸⁰, ⁷⁸⁹AA⁷⁹⁰, F794A, R801A, Q807E, A808S, and D815A) along with wild-type eEF2 and negative control (enhanced green fluorescent protein, EGFP), were then screened for interaction with CPEB2 by co-immunoprecipitation (co-IP). This assay identified F794 as a critical eEF2 residue for binding to CPEB2. To confirm the initial screening result, we selected 3 mutants (⁷⁷⁸AAA⁷⁸⁰, F794A, and D815A designated as mut1, mut2, and mut3) for further analysis (**Figure 2a**). The interaction of wild-type and mutant eEF2 with CPEB2 was analyzed again by reciprocal co-IP. The results in **Figure 2b** show that the F794A mutation (mut2) abolished binding to CPEB2.

Next, we investigated whether disrupting the association between CPEB2 and eEF2 affects HIF-1α expression *in vivo*. Because eEF2 is an essential and abundant translational factor, its ectopic expression alone was insufficient to override the function of endogenous eEF2. Thus, we tested the F794A mutant under knockdown of endogenous eEF2 condition. HeLa cells were transfected with plasmids lacking (siCtrl) or containing a short hairpin sequence for human eEF2 (siheEF2) and

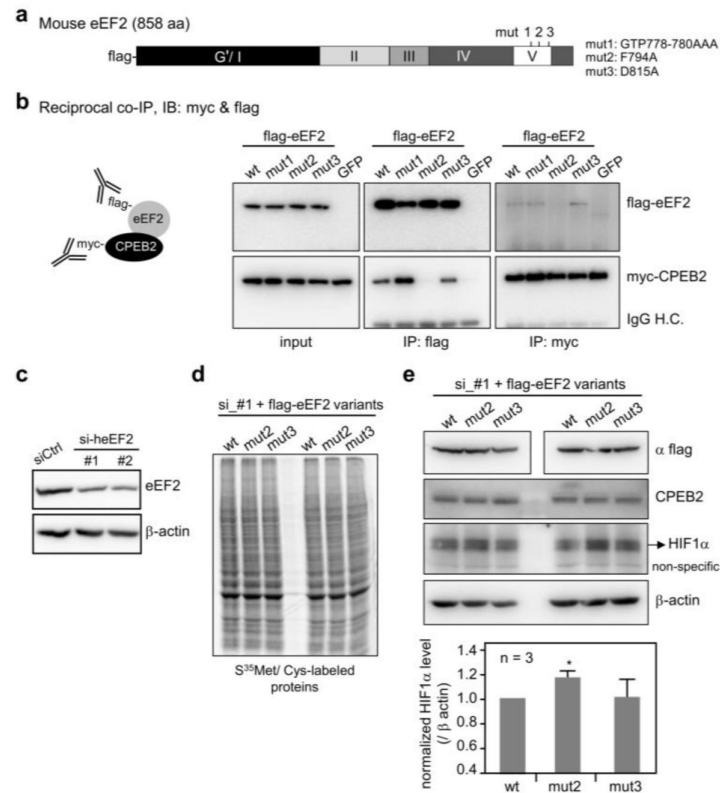


Figure 2.

Evaluation of the predicted CPEB2-interacting amino acid residues in eEF2. **(a)** Salient features of mouse eEF2, showing the various domains and the mutated amino acids in domain V. mut 1, G778A, T779A, and P780A; mut 2, F794A; mut 3, D815A. **(b)** Reciprocal co-IP. The 293T cells expressing myc-CPEB2 along with wt or mutant flag-eEF2 or control GFP were harvested and then precipitated with flag or myc IgG. The precipitated substances were used for western blotting with myc and flag antibodies. IP, immunoprecipitation; IB, immunoblotting; IgG H.C., IgG heavy chain. **(c)** HeLa cells transfected with the plasmid expressing shRNA against human eEF2 (siheEF2) were harvested after 4-day puromycin selection for western blotting. HeLa cells transfected with the eEF2 knockdown plasmid along with flag-tagged wt or mutant mouse eEF2 after 4-day selection with puromycin and G418 were used for **(d)** S³⁵Met/cys-labeling of synthesized proteins or **(e)** western blotting with the denoted antibodies. The normalized HIF-1α protein level (HIF-1α/β-actin signal) was calculated and expressed as mean ± SEM from three independent experiments. 2-tailed Student's t-test, * < 0.05.

subjected to puromycin selection. Both shRNA sequences, specifically knocking down human but not mouse eEF2, decreased endogenous eEF2 after 4 days (**Figure 2c**). HeLa cells were then transfected with the siheEF2 and flag-meEF2 (wild-type, mut2, or mut3) plasmids and subjected to puromycin and G418 selection for four days. Cells that survived were incubated with S^{35} -methionine/cysteine to metabolically label synthesized proteins. The expression of the F794A or D815A mutant did not affect general protein synthesis (**Figure 2d**). However, the level of HIF-1 α , but not CPEB2 or β -actin, was selectively increased in HeLa cells reconstituted with the F794A mutant (**Figure 2e**). **Supplementary Figure S1** shows the uncropped immunoblot images. Thus, the eEF2 F794A mutation influences the translation of CPEB2-targeted HIF-1 α mRNA without affecting general translation function.

Discussion

Identifying PPI-hot spots is challenging especially when the complex structure is lacking. A key hurdle is the lack of experimental data on PPI-hot spots, which hampers the training of accurate machine-learning models for their prediction. Here, we have introduced two novel elements that have helped to identify PPI-hot spots using the unbound structure. First, we have constructed a dataset comprising 414 experimentally known PPI-hot spots and 504 nonhot spots, and *carefully* checked that PPI-hot spots have no mutations resulting in $\Delta\Delta G^{\text{bind}} < 0.5$ kcal/mol, whereas nonhot spots have no mutations resulting in $\Delta\Delta G^{\text{bind}} \geq 0.5$ kcal/mol or impact binding in immunoprecipitation or GST pull-down assays (see **Methods**). In contrast, SPOTONE⁵³ employed nonhot spots defined as residues that upon alanine mutation resulted in $\Delta\Delta G^{\text{bind}} < 2.0$ kcal/mol. Notably, previous PPI-hot spot prediction methods did not employ PPI-hot spots whose mutations have been curated to significantly impair/disrupt PPIs in UniProtKB (see **Introduction**). Second, we have introduced novel features derived from *unbound* protein structures such as the gas-phase energy of the target protein relative to its unfolded state. The importance test results indicated the gas-phase energy as an important feature. This finding can be rationalized by considering how PPI-hot spots make significant contributions to the overall binding free energy, ΔG^{bind} . PPI-hot spots can enhance favorable enthalpic contributions to the ΔG^{bind} through hydrogen bonds or vdW contacts across the protein's interface. This makes them energetically *unstable* in the absence of the protein's binding partner and solvent, hence the gas-phase energy was found to be an important input feature. Alternatively, PPI-hot spots can counteract unfavorable entropic loss upon protein binding by maintaining an optimal binding scaffold, hence they are energetically *stable*.

Methods that rely on complex structures generally predict residues that make multiple contacts across the protein-protein interface as PPI-hot spots. Some of these methods assume that PPI-hot spots are exclusively located at the interface and aim to spot them among the interface residues.⁴⁰ In contrast, PPI-Hotspot^{ID} leverages evolutionary conservation, residue type, and stability principles based on the free protein structure to detect PPI-hot spots, including those lacking direct contact with the partner protein. Such noninterface PPI-hot spots may serve to maintain an optimal scaffold for protein binding, and are not uncommon: from our analysis of the 243 true PPI-hot spots in proteins with the complex structures, we found 67 “noninterface” PPI-hot spots with no hydrogen bonds and/or vdW contacts across the protein interface. PPI-Hotspot^{ID} identified 43 of these 67 noninterface PPI-hot spots. An illustrative

example is the binding of Golgi resident protein (GCP60) with phosphatidylinositol 4-kinase β (PI4K— β). PPI-Hotspot^{ID} correctly predicted all 4 experimentally known GCP60 PPI-hot spots including F19 and Y46, which do not form hydrogen bonds across the interface with PI4K— β (**Figure 3**). These results highlight the ability of PPI-Hotspot^{ID} to identify PPI-hot spots involved in indirect interactions with partner proteins.

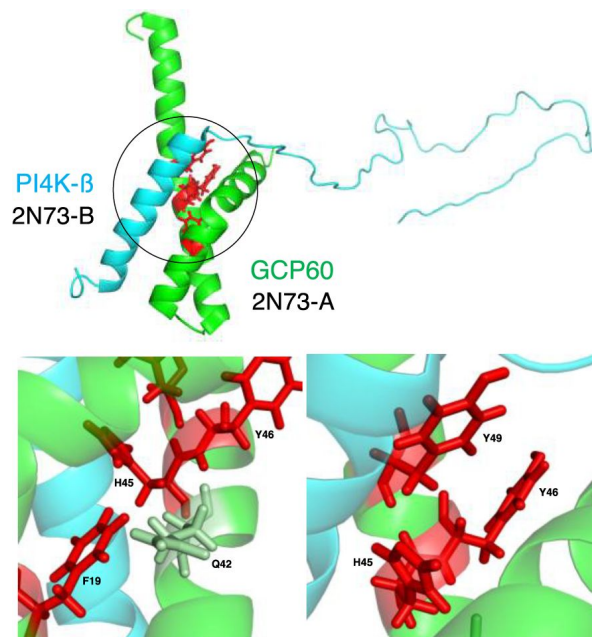


Figure 3.

(Top) The structure (PDB 2N73)⁸⁴ of GCP60 (green) in complex with PI4K-β (cyan) with the GCP60—PI4K-β interface encircled. (Bottom) The four experimentally known PPI-hot spots of GCP60 are shown in red. H45 and Y49 form hydrogen bonds across the interface with PI4K-β. Although F19 and Y46 do not directly contact PI4K-β, F19 is in vdW contact with Q42, which in turn forms vdW contacts with H45, whereas Y46 is in vdW contact with both H45 and Y49.

Proteins typically interact with multiple partners, but their PPI-hot spots may have been experimentally characterized for only a few partners. In some cases where PPI-Hotspot^{ID} predicted residues that were absent in the PPI-Hotspot+PDB^{BM(1.1)} as PPI-hot spots, the protein's complex structures with other binding partners show intermolecular hydrogen bonds between PPI-Hotspot^{ID}-predicted residues and residues of the respective partner proteins. This suggests that some of the PPI-Hotspot^{ID}-predicted residues might be potential PPI-hot spots for other binding partners. For example, the death domain of CRADD (caspase-recruitment domain and death domain-containing adaptor protein) contains 7 experimentally known PPI-hot spots (N121, Q125, Y146, R147, K149, V156, Q169) critical for its interaction with PIDD (p53-induced death domain-containing protein). Based on the free crystal structure of CRADD (PDB 2O71-A),⁶⁷ PPI-Hotspot^{ID} correctly predicted 3 true positives (Y146, R147, and Q169) as well as G128. In the oligomeric structure (PDB 2OF5)⁶⁸ of 7 CRADD proteins in complex with 5 PIDD proteins, G128 shows no hydrogen-bonding interactions, but its neighbor, L127, forms a backbone---side chain hydrogen bond with R147 in another CRADD chain (**Figure 4**). A positively charged G128R mutation would repel the nearby positively charged R147 in another CRADD chain, thus disrupting the CRADD---CRADD interface and decreasing CRADD's affinity for PIDD. Experimental data showed that the G128R CRADD mutant did not co-immunoprecipitate the PIDD death domain, and patients who have non-syndromic mental retardation possess the G128R mutant.⁶⁹ Thus, PPI-Hotspot^{ID} could unveil a PPI-hot spot, G128, that is not apparent from the 2OF5 complex structure: although G128 does not directly interact with PIDD, its mutation, especially to an Arg, might perturb the CRADD---CRADD interface and thus CRADD's oligomeric structure and binding affinity for PIDD.

The ability of PPI-Hotspot^{ID} to detect PPI-hot spots provides biologists with a useful tool, as alanine-scanning mutagenesis and protein-protein complex structure determination to identify PPI-hot spots are laborious, time-consuming, and costly. Conventional methods based on complex structures might miss nonobvious PPI-hot spots with no direct interactions with the protein's partner. AlphaFold-multimer and future improved protein-protein complex prediction methods require knowledge of interacting partners and independent calculations for each known partner, which reduces the overall efficiency. Moreover, solved/modeled proteinprotein complex structures only reveal the interface residues. In contrast, PPI-Hotspot^{ID} can reveal nonobvious PPI-hot spots as well as potential PPI-hot spots for other protein partners, thus helping to elucidate the different PPI mechanisms.

Methods

Dataset. True PPI-hot spots.

We updated the PPI-Hotspot+PDB^{BM} benchmark by removing 2 fused protein structures and adding new PPI-hot spots by (i) reviewing references in ASEdb⁵⁵ to include nonalanine mutations with $\Delta\Delta G^{\text{bind}} > 2$ kcal/mol, and (ii) checking the experimental data of certain mutations in UniProtKB.⁵⁸ For example, the PPI-Hotspot+PDB^{BM} benchmark included R43A in aprataxin (UniProtID Q7Z2E3), annotated as “loss of interaction with MDC1”, but not K52A, annotated as “impairs interaction with MDC1”. However, when we checked the experimental data in the UniProtKB reference, the binding bands were absent for both R43A and K52A mutants, therefore we added K52A as a PPI-hot spot. The updated benchmark, termed PPI-Hotspot+PDB^{BM(1.1)}, contains 414 PPI-hot spots. Among these, 104 PPI-hot spots in 32 nonredundant proteins are based on mutations resulting in $\Delta\Delta G^{\text{bind}} \geq 2$ kcal/mol from ASEdb⁵⁵ and SKEMPI2.0⁵⁶ with no known mutations resulting in $\Delta\Delta G^{\text{bind}} < 0.5$ kcal/mol. The remaining 310 PPI-hot spots in 128 nonredundant proteins are based on mutations that are manually curated in UniProtKB⁵⁸ to significantly impair/disrupt PPIs. Two of the proteins have PPI-hot spots from ASEdb/SKEMPI2.0 and UniProtKB, resulting in a total of 158 nonredundant proteins with free structures harboring 414 PPI-hot spots (Supplementary Tables S1 and S2).

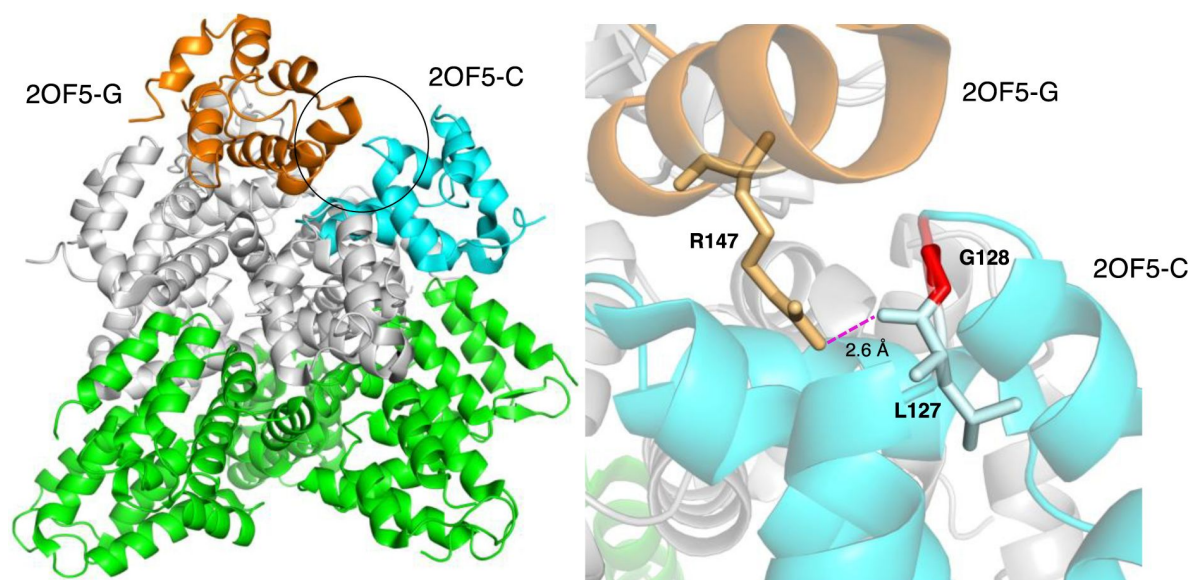


Figure 4.

(Left) The structure (PDB 2OF5)⁶⁸ of 7 CRADD proteins in complex with 5 PIDD proteins. The circle shows the CRADD—CRADD interface between chains C (cyan) and G (orange), whereas the other 5 CRADD chains are in gray, and the 5 PIDD proteins are in green. (Right) G128 (red) in CRADD (chain C) participates indirectly in CRADD—CRADD interactions via a backbone—side chain hydrogen bond between its neighbor, L127, and R147 in another CRADD (chain G).

True PPI-nonhot spots.

To obtain PPI-nonhot spots for the 158 nonredundant proteins with true PPI-hot spots, we identified residues from ASEdb⁵⁵ and SKEMPI2.0⁵⁶ databases where mutations to alanine/nonalanine resulted in protein $\Delta\Delta G^{\text{bind}} < 0.5$ kcal/mol. We also identified residues in the UniProtKB where mutations to alanine/nonalanine were curated *not* to perturb PPIs. We *manually* checked each reference to ensure that mutations of these residues did not lead to $\Delta\Delta G^{\text{bind}}$ changes ≥ 0.5 kcal/mol or impact binding in immunoprecipitation or GST pull-down assays. PPI-nonhot spots in non-native proteins or regions with missing structures were excluded. [Supplementary Table S1](#) lists the 504 PPI-nonhot spots found in 75 proteins with free structures.

Input features.

To distinguish PPI-hot spots from PPI-nonhot spots, we input sequence, structural, and stability features of each residue in the protein for training various machine learning classifiers. The input features for each residue i of a protein included its aa type, conservation score, secondary structure, SASA, gas-phase energy, and respective components, polar solvation free energy, and nonpolar solvation free energy. The secondary structure, SASA, and energy components of each residue were computed using the DSSP program,⁷⁰ FreeSasa⁷¹ and AmberTools version 20,⁷² respectively, using default parameters.

Per-Residue Free Energy Contributions.

For a given free protein structure, the Reduce program⁷³ was used to add hydrogens and assign the protonation states of ionizable residues. Additional missing heavy and hydrogen atoms were added using the AmberTools version 20⁷² and the Amber FF19SB forcefield.⁷⁴ To eliminate any steric clashes, we performed a conjugate gradients minimization with constraints on the heavy atoms using the Generalized Born model for 500 steps. The resulting structure was used to compute the per-residue energy/free energy contributions using the MMPBSA (Molecular Mechanics Poisson-Boltzmann Surface Area) module in AmberTools.⁷² For each residue i in the protein, we computed the (i) molecular mechanics energy

$E_i^{\text{gas}} = E_i^{\text{MM,int}} + E_i^{\text{MM,vdW}} + E_i^{\text{MM,ele}}$, where $E_i^{\text{MM,int}}$ includes contributions from bonded terms, $E_i^{\text{MM,vdW}}$ is the vdW interaction energy, and $E_i^{\text{MM,ele}}$ is the electrostatic interaction energy as well as (ii) the polar ($\Delta G_i^{\text{solv,pol}}$) and nonpolar ($\Delta G_i^{\text{solv,npl}}$)

solvation free energies relative to the corresponding values of residue i in an extended reference state where the residues do not interact with one another.⁷⁵

Per-Residue Conservation Score.

To calculate the conservation score, k_i^C , of residue i in a protein, we implemented a method similar to ConSurf^{76,77} to run in parallel with the energy evaluation code. First, we searched the UNIREF-90 database⁷⁸ using HMMER⁷⁹ to find sequences similar to the target sequence. Near-duplicates were removed by clustering matched sequences with $\geq 95\%$ pairwise sequence identity using CD-hit⁸⁰ and keeping only one representative. Since HMMER⁷⁹ may only find good matches for a small proportion of the target sequence, we compared the HMMER sequences with the target sequence. We kept only those with $> 60\%$ overlap with the target sequence, and discarded sequences that were dissimilar ($\leq 35\%$ sequence identity) or nearly identical ($\geq 95\%$ sequence identity). Next, we pairwise aligned the remaining sequences, and if two sequences overlapped by $> 10\%$ of the sequence, we rejected the shorter sequence. After this filtering process, the resulting HMMER hits were used or if the number of hits exceeded 300, we selected the top 300 hits. These sequences were then aligned to the target sequence using MAFFT-LINSi.⁸¹ We then used the Rate4Site program⁸² to compute position-specific evolutionary rates from the

generated multiple sequence alignment. These rates were normalized and grouped into ConSurf grades ranging from 1 to 9, where $k^C = 1$ represents the most rapidly evolving residues, and $k^C = 9$ indicates the most conserved residues.

Generating PPI-hot spot Predictive Model using AutoGluon.

We provided all the aforementioned residue features including the conservation score, k_i^C , aa type, DSSP secondary structure, SASA_i, $E_i^{MM,int}$, $E_i^{MM,vdW}$, $E_i^{MM,ele}$, E_i^{MM} , $\Delta G_i^{solv,pol}$, and $\Delta G_i^{solv,npl}$ to the Tabular module in AutoGluon v0.8.2 (<https://auto.gluon.ai/stable/index.html>).

AutoGluon was chosen for model training and validation due to its robustness and user-friendly interface, allowing for the simultaneous and automated exploration of various machine-learning approaches and their combinations. Instead of using a single training set to train the model and a separate test set to evaluate its performance, we employed cross-validation, as it utilizes the entire dataset for both training and testing, making efficient use of the limited data on PPI-hot spots and PPI-nonhot spots. AutoGluon-Tabular automatically chose a random partitioning of our dataset into multiple subsets/folds for training and validation. Notably, the training and validation data share insignificant homology, as the average pairwise sequence identity in our dataset is 26%. Each fold was used once as a test set, while the remaining folds served as the training set. For each test set, the model's performance was measured using the F1 score.

AutoGluon then trained individual “base” models, including LightGBM, CatBoost, XGBoost, random forests, extremely randomized trees, neural networks, and K-nearest neighbours. Using the aggregated predictions of the base models as features in addition to the original features, AutoGluon trained multiple “stacker” models, whose predictions were fed as inputs to additional higher layer stacker models in an iterative process called multi-layer stacking. The output layer used ensemble selection to aggregate the predictions of the stacker models. To improve stacking performance, AutoGluon used all the data for both training and validation through repeated k -fold bagging of all models at all layers of the stack, where k is determined by best precision. We refer the reader to the original study by Erikson et al.⁸³ which provides details on the methodology including the types of “base” models, multi-layer stack ensembling, and repeated k -fold bagging. Based on the highest mean F1 score, AutoGluon yielded a final PPI-hot spot predictive model that is a weighted regularized ensemble comprising more than a dozen different models (<https://auto.gluon.ai/dev/api/autogluon.tabular.models.html>).

Selecting Key Features.

Next, we evaluated the importance of each feature by performing a permutation-based test (part of the AutoGluon package), in which a feature in a column was randomly shuffled across different residues (rows), and the F1 score was evaluated. The importance test results revealed the four most important residue features, which in order of their importance, are (i) k^C , (ii) aa residue type, (iii) SASA_i, and (iv) ΔG_i^{gas} . These four features were used to train an ensemble of machine-

learning models using the entire data set, consisting of 414 true PPI-hot spots and 504 nonhot spots. The resulting PPI-hot spot prediction model, named PPI-Hotspot^{ID}, yielded a F1 score comparable to the F1 score obtained using the initial set of 10 features. PPI-Hotspot^{ID} was implemented as a freely accessible web server (<https://github.com/wrigjz/ppihotspotid>) with access to 4 virtual CPUs and 8 GB of memory. Calculations for a 539-residue protein (PDB 1c2bA) took 35 minutes.

Detecting PPI-hot spots Using the AlphaFold-Multimer-predicted Interface.

In cases where experimental complex structures are unavailable, can the protein–protein complexes modeled by AlphaFold-Multimer⁶⁰ be used to identify PPI-hot spots using the predicted interface residues? To address this, we first identified PPI-hot spots within the PPI-Hotspot+PDB^{BM(1.1)} dataset that lack experimentally determined protein complex structures. Not all the 414 PPI-hot spots in the PPI-Hotspot+PDB^{BM(1.1)} have sequence information and thus UniProtID of the respective binding partners (see Supplementary Table S1), leaving 360 PPI-hot spots in 135 proteins associated with 155 pairs of PPIs, as some proteins are involved in multiple PPIs. Ninety of the 155 PPI pairs have complex structures in the PDB. For the 65 PPI pairs lacking complex structures, 17 pairs contain > 1,100 residues, exceeding the current size limit of AlphaFold-Multimer⁶⁰. Thus, we generated structural models for the remaining 48 complexes using the AlphaFold-Multimer module in the ColabFold version 1.3.0⁶⁴ with default settings. For each AB complex, the input sequence for protein A was based on the free structure sequence in the PPI-Hotspot+PDB^{BM(1.1)}, whereas that for protein B was retrieved in its entirety from the UniProtKB,⁵⁸ as the binding region in protein B was unknown. Based on the AMBER-relaxed model structure with the highest pTM score, interface residues were defined as residues of protein A with ≥1 atom within a 5 Å cutoff of any protein B atom.

Experimental Verification of Predicted eEF2 PPI-hot spots. Plasmid construction.

The predicted PPI-hot spots and PPI-*non*hot spots were mutated by QuikChange Site-Directed Mutagenesis Kit (Stratagene). The pcDNA3.1-flag-meEF2 plasmid was used as the PCR template and the sets of sense and antisense primers for mutagenesis are listed in **Supplementary Figure S2**. All constructs were sequenced to confirm the mutations. The shRNA clones, #1 TRCN0000047908 (GCGATCATGAATTTCAAGAAA) and #2: TRCN0000047910 (GCAGTACCTCAACGAGATCAA) against human eEF2 mRNA were obtained from the RNAi Core Facility (Academia Sinica).

Testing eEF2-CPEB2 interactions using co-immunoprecipitation (co-IP) and reciprocal co-IP.

HEK-293T cells obtained from American Type Culture Collection (ATCC, # CRL-3216) were cultured in DMEM with 10% fetal bovine serum (FBS). For reciprocal co-IP, the 8 µg DNA mixture containing 3 µg myc-CPEB2 and 5 µg flag-eEF2 (or a negative control, GFP) plasmids was transfected into a 10-cm dish of 293T cells using lipofectamine 2000. We transfected more flag-eEF2 plasmid DNA than myc-CPEB2 plasmid because myc-CPEB2 is expressed more abundantly than flag-eEF2. Overnight transfected cells were lysed in 500 µl IP buffer (20 mM Hepes, pH 7.4, 100 mM NaCl, 1 mM MgCl₂, 0.1% TritonX-100, 10% glycerol, 0.5 mM DTT, 1X protease inhibitor cocktail and 100 µg/ml RNaseA) and centrifuged at 10,000 *xg* for 3 min at 4°C. The supernatant was equally divided and incubated with Protein G beads bound with myc or flag antibody for 3 hours at 4°C to respectively pull down myc-CPEB2 and flag-eEF2. The beads were washed five times with 300 µl IP buffer. If myc-CPEB2 and flag-eEF2 interact, myc-CPEB2 can co-precipitate with flag-eEF2 on flag antibody-beads, whereas flag-eEF2 can co-precipitate with myc-CPEB2 on myc antibodybeads. GFP was used as a negative control to ensure that the signals on the beads were caused by binding between flag-eEF2 and myc-CPEB2. The precipitated proteins were separated on a sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) for western blot analysis. Similarly, for the initial co-IP screening, the 4 µg DNA mixture containing 1.5 µg myc-CPEB2 and 2.5 µg flag-eEF2 (or a negative control, EGFP) plasmids was transfected into a 6-cm dish of 293T cells, harvested in 200 µl IP buffer and immunoprecipitated by using flag antibody-bound beads.

Functional impact of eEF2 mutants on HIF-1a and global protein synthesis.

HeLa cells obtained from ATCC (# CCL-2) were cultured in DMEM with 10% FBS. Each 6-cm plate of HeLa cells was transfected with 2 µg human eEF2 knockdown plasmid and 2 µg flag-meEF2 wild-type/mutant plasmid. Overnight transfected cells were selected with 0.5 µg/ml puromycin and 600 µg/ml G418 for three days to knock down endogenous eEF2 and maintain the expression of flag-meEF2, respectively. The selected cells were incubated with 10 µM MG132 for 4 hours before harvesting for western blotting of HIF-1a or replaced with 2 ml Met/Cys-lacking DMEM with 1% FBS and 60 µCi ³⁵S-Met/Cys (Perkin Elmer, cat# NEG772002MC) for 2 hours before separation on an SDS-PAGE.

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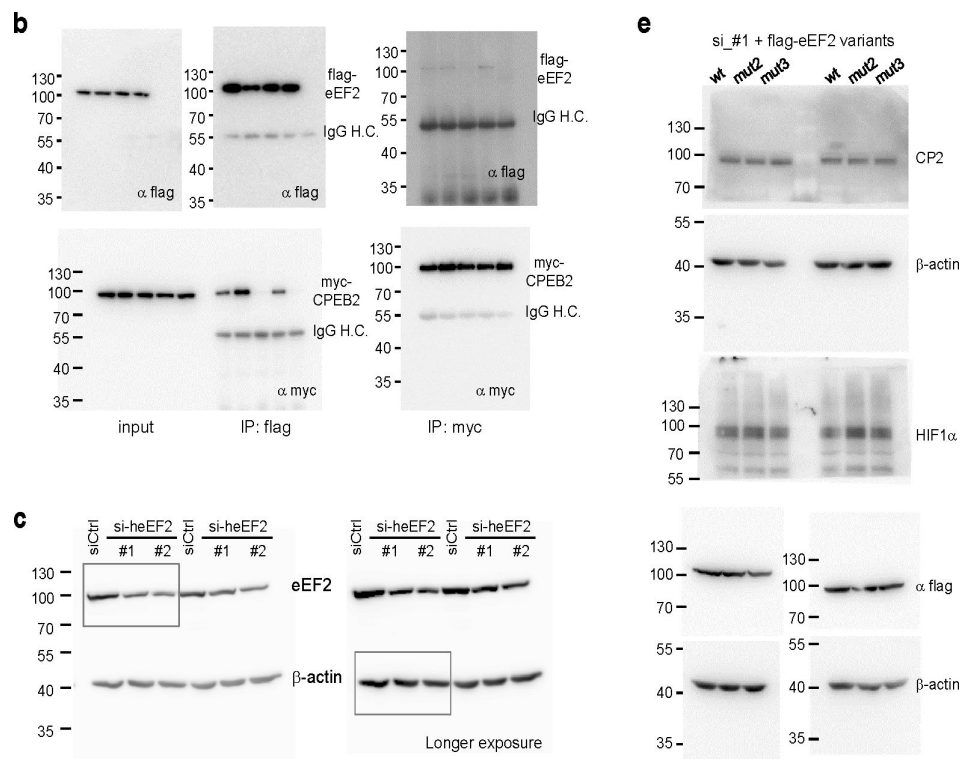
Competing interests

The authors declare no competing interests.

Additional Declarations:

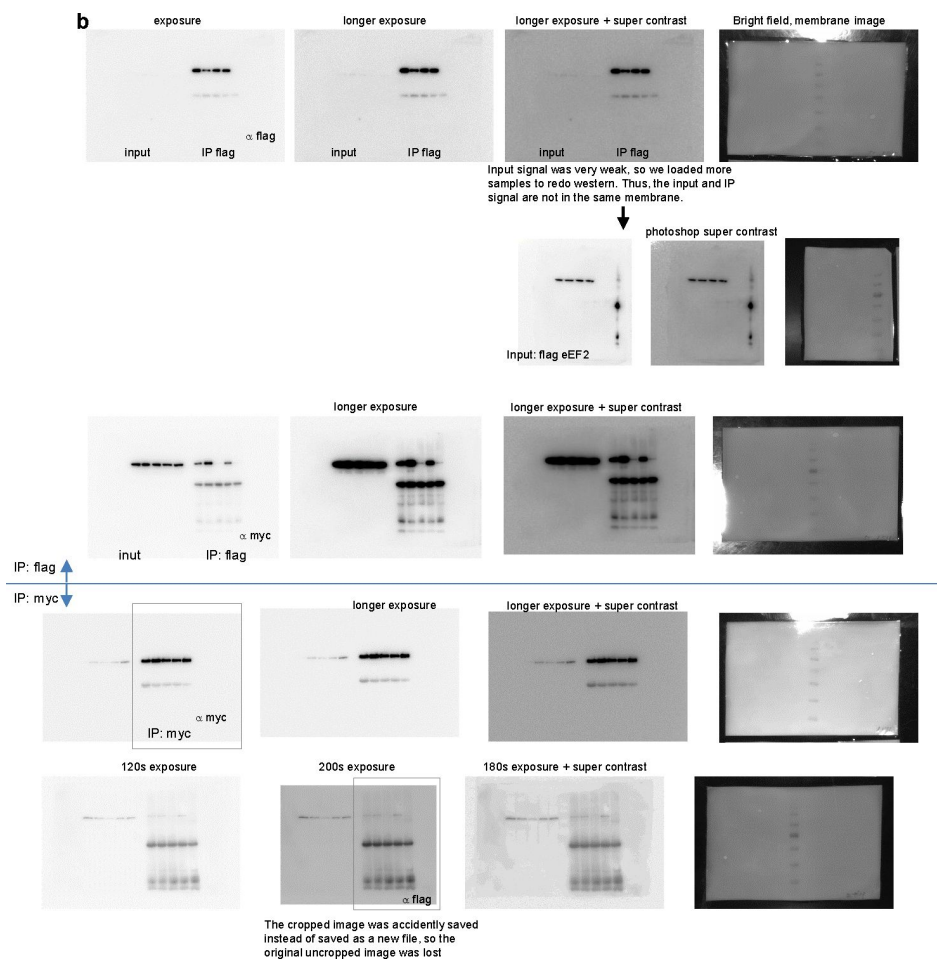
The authors declare no competing interests.

Supplementary Information



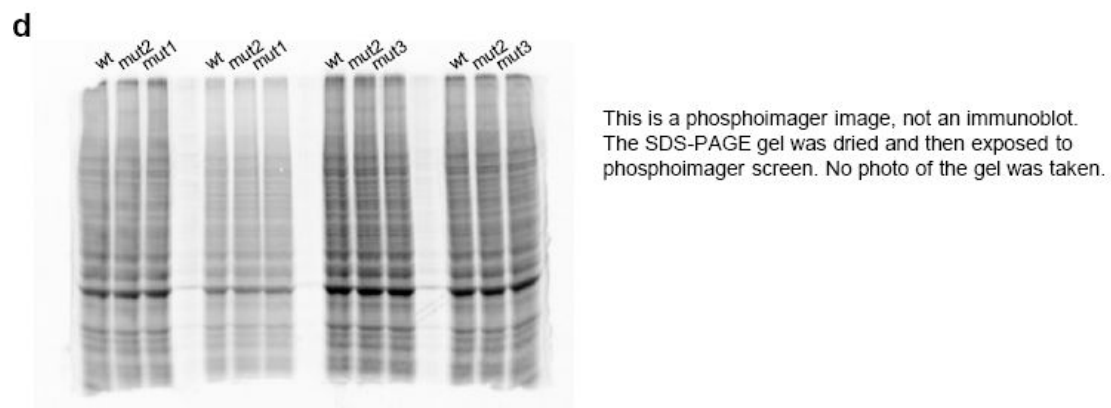
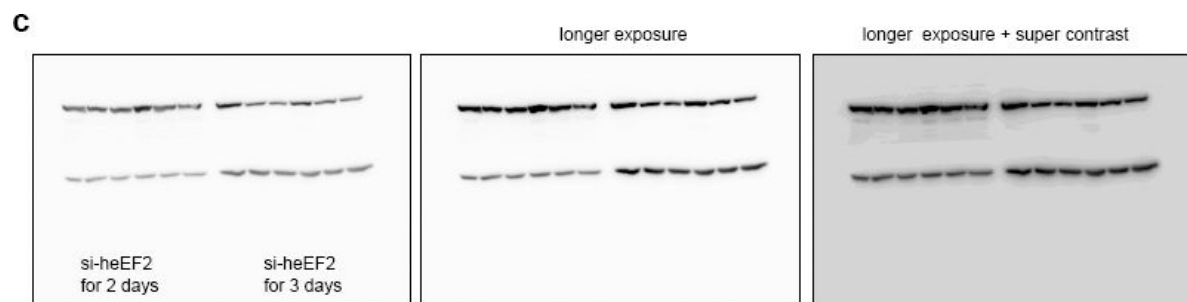
Supplementary Figure S1a.

Uncropped immunoblot images. The uncropped images for Figures 2b [\[link\]](#), 2c [\[link\]](#), and 2e [\[link\]](#) are shown with indicated molecular weight marker.



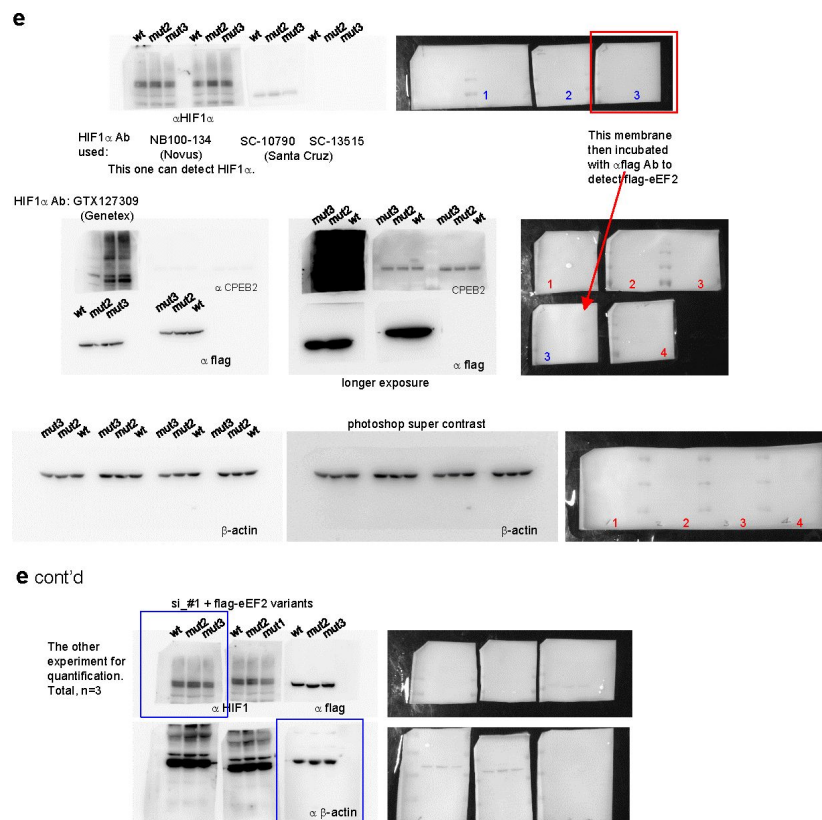
Supplementary Figure S1b.

Uncropped immunoblot images of the entire membranes for Figure 2b [↗](#).



Supplementary Figure S1c.

Uncropped images of the entire membrane for Figure 2c [↗](#) and phosphoimager file for 2d.



Supplementary Figure 1d.

Uncropped immunoblot images of the cut membranes for Figure 2e [↗](#).

Mutated sites	Forward (sense)	Reverse (antisense)
G778A/ T779A/ P780A	GAGGAGTCCCAGGTGGCTGCTGCTG CCATGTTTGTGGTCAAGGCAT	ATGCCTTGACCACAAACATGGCAGCAG CAGCCACCTGGGACTCCTC
F794A	TGCCTGTCAATGAGTCCGCCGGCTTC ACCGCTGATCT	AGATCAGCGGTGAAGCCGGCGGACTCA TTGACAGGCA
D815A	CCCCCAGTGCGTGTTTGCCCACTGGC AGATCCTGC	GCAGGATCTGCCAGTGGGCAAACACGC ACTGGGG
L763A	GTGGCATCTACGGTGTGCGAACAG GAAGCGTGGCC	GGCCACGCTTCCTGTTCGCGACACCGT AGATGCCAC
K766A/ R767A/ G768A	CTACGGTGTCTGAACAGGGCAGCTG CCCATGTGTTTGAGGAGTCC	GGACTCCTCAAACACATGGGCAGCTGC CCTGTTCAAGGACACCGTAG
E773Q	CGTGGCCATGTGTTTGAGCAGTCCCA GGTGGCTGGTA	TACCAGCCACCTGGGACTGCTCAAACA CATGGCCACG
P789A/ V790A	TGTGGTCAAGGCATACCTGGCTGCTA ATGAGTCCTTTGGCTTCAC	GTGAAGCCAAAGGACTCATTAGCAGCC AGGTATGCCTTGACCACA
R801A	GGCTTCACCGCTGATCTGGCATCCAA CACCGGCGGCCA	TGGCCGCCGGTGTGGATGCCAGATCA GCGGTGAAGCC
Q807E	GATCCAACACCGCGGCGAGGCCTT CCCCCAGTGCGTGTT	AACACGCACTGGGGGAAGGCCTCGCC GCCGGTGTGGATC
A808S	AACACCGGCGGCCAGAGCTTCCCCC AGTGCCTGT	ACACGCACTGGGGGAAGCTCTGGCCGC CGGTGTT

Supplementary Figure S2.

Primer sequences. The sequences of sense (forward) and antisense (reverse) primers used for site-directed mutagenesis.

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The paper describes a program developed to identify PPI-hot spots using the free protein structure and compares it to FTMap and SPOTONE, two webserver that they consider as competitive approaches to the problem. We appreciate the effort in providing a new webserver that can be tested by the community but we continue to have major concerns:

(1) The comparison to the FTMap program is problematic. The authors misinterpret the article they refer to, i.e., Zerbe et al. "Relationship between hot spot residues and ligand binding hot spots in protein-protein interfaces" *J. Chem. Inf. Model.* 52, 2236-2244, (2012). FTMap identifies hot spots that bind small molecular ligands. The Zerbe et al. article shows that such hot spots tend to interact with hot spot residues on the partner protein in a protein-protein complex (emphasis on "partner"). Thus, the hot spots identified by FTMap are not the hot spots defined by the authors. In fact, because the Zerbe paper considers the partner protein in a complex, the results cannot be compared to the results of Chen et al. This difference is missed by the authors, and hence the comparison of the FTMap is invalid.

(2) Chen et al. use a number of usual features in a variety of simple machine-learning methods to identify hot spot residues. This approach has been used in the literature for more than a decade. Although the authors say that they were able to find only FTMap and SPOTONE as servers, there are dozens of papers that describe such a methodology. Some examples are given here: (Higa and Tozzi, 2009; Keskin, et al., 2005; Lise, et al., 2011; Tuncbag, et al., 2009; Xia, et al., 2010). There are certainly more papers. Thus, while the web server is a potentially useful contribution, the paper does not provide a fundamentally novel approach.

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Author response:

The following is the authors' response to the original reviews.

eLife assessment

The manuscript presents a machine-learning method to predict protein hotspot residues. The validation is incomplete, along with the misinterpretation of the results with other current methods like FTMap.

We believe that validation is complete: The two most common techniques for testing and validating machine-learning methods are to split the dataset into either (1) a training set and a test set with a fixed ratio (e.g., 70% for training and 30% for testing) or (2) multiple subsets/folds; i.e., cross-validation. We did not employ a training set to train the model and a

separate test set to evaluate its performance, as Reviewer 2 assumed. Instead, we employed cross-validation, as it helps reduce the variability in performance estimates compared to a single training/test split, and utilizes the entire dataset for training and testing, making efficient use of the limited data. Each fold was used once as a test set and the remaining folds as the training set - this process was repeated for each fold and the model's performance was measured using the F1 score. We had listed the mean validation F1 score in Table 1.

We have clarified our comparison with FTMAP - see reply to point 1 of reviewer 1 below.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The paper describes a program developed to identify PPI-hot spots using the free protein structure and compares it to FTMap and SPOTONE, two webserver that they consider as competitive approaches to the problem. On the positive side, I appreciate the effort in providing a new webserver that can be tested by the community but have two major concerns as follows.

(1) The comparison to the FTMap program is wrong. The authors misinterpret the article they refer to, i.e., Zerbe et al. "Relationship between hot spot residues and ligand binding hot spots in protein-protein interfaces" J. Chem. Inf. Model. 52, 2236-2244, (2012). FTMap identifies hot spots that bind small molecular ligands. The Zerbe et al. article shows that such hot spots tend to interact with hot spot residues on the partner protein in a protein-protein complex (emphasis on "partner"). Thus, the hot spots identified by FTMap are not the hot spots defined by the authors. In fact, because the Zerbe paper considers the partner protein in a complex, the results cannot be compared to the results of Chen et al. This difference is missed by the authors, and hence the comparison of the FTMap is invalid. I did not investigate the comparison to SPOTONE, and hence have no opinion.

Brenke et al. (Bioinformatics 2009 25: 621-627), who developed FTMAP, defined hot spots as regions of the binding surface that “contribute a disproportionate amount to the binding free energy”. Kozakov et al. (Proc. Natl. Acad. Sci. 2011:108, 13528-13533) used unbound protein structures as input to FTMap to predict binding hot spots for protein-protein interactions (PPIs), which are defined as regions (so-called consensus sites) on a protein surface that bind multiple probe clusters – the main hot spot is the largest consensus site binding the largest number of probe clusters.

Zerbe et al. (J. Chem. Inf. Model. 2012:52, 2236) noted that a consensus “site is expected to be important in any interaction that involves that region of the target independent of any partner protein.” They showed that for hot spot residues found by Ala scanning not only overlapped with the probe ligands but also form consensus sites, as shown in Figure 4. They stated that “A residue can also be identified as a hot spot by alanine scanning if it contributes to creating such a favorable binding environment by being among the residues forming a consensus site on the protein to which it belongs.”

To clarify the comparison with FTmap in the revised version, we have added the following sentence in the Abstract on p. 3:

“We explored the possibility of detecting PPI-hot spots using (i) FTMap in the PPI mode, which identifies hot spots on protein-protein interfaces from the free protein structure, and (ii) the interface residues predicted by AlphaFold-Multimer.”

We have added the following sentences in the Introduction section on p. 4:

“We explored the possibility of detecting PPI-hot spots using the FTMap server in the PPI mode, which identifies hot spots on protein-protein interfaces from free protein structures.⁴⁵ These hot spots are identified by consensus sites – regions that bind multiple probe clusters.^{42,45,59} Such regions are deemed to be important for any interaction involving that region of the target, independent of partner protein.⁴² PPIhot spots were identified as residues in van der Waals (vdW) contact with probe ligands within the largest consensus site containing the most probe clusters.”

and in the Results section on p. 5:

“Given the free protein structure, PPI-HotspotID and SPOTONE⁵³ predict PPI-hot spots based on a probability threshold (> 0.5). FTMap, in the PPI mode, detects PPIhot spots as consensus sites/regions on the protein surface that bind multiple probe clusters.⁵⁹ Residues in vdW contact with probe molecules within the largest consensus site were compared with PPI-hotspotID/SPOTONE predictions.”

(2) Chen et al. use a number of usual features in a variety of simple machine-learning methods to identify hot spot residues. This approach has been used in the literature for more than a decade. Although the authors say that they were able to find only FTMap and SPOTONE as servers, there are dozens of papers that describe such a methodology. Some examples are given here: (Higa and Tozzi, 2009; Keskin, et al., 2005; Lise, et al., 2011; Tuncbag, et al., 2009; Xia, et al., 2010). There are certainly more papers. Thus, while I consider the web server as a potentially useful contribution, the paper does not provide a fundamentally novel approach.

Our paper introduces several novel elements in our approach:

(1) Most PPI-hot spot prediction methods employ PPI-hotspots where mutations decrease protein binding free energy by > 2 kcal/mol (J. Chem. Inf. Model. 2022, 62, 1052). In contrast, our method incorporates not only PPI-hot spots with such binding free energy changes, but also those whose mutations have been curated in UniProtKB to significantly impair/disrupt PPIs. Because our method employs the largest collection of experimentally determined PPI-hot spots, it could uncover elusive PPI-hot spots not within binding interfaces, as well as potential PPI-hot spots for other protein partners (see point 3 below).

(2) Whereas most machine-learning methods for PPI-hot spot prediction focus on features derived from (i) primary sequences or (ii) protein-protein complexes, we introduce novel features such as per-residue free energy contributions derived from unbound protein structures. We further revealed the importance of one of our novel features, namely, the gas-phase energy of the target protein relative to its unfolded state and provided the physical basis for its importance. For example, PPI-hot spots can enhance favorable enthalpic contributions to the binding free energy through hydrogen bonds or van der Waals contacts across the protein’s interface. This makes them energetically unstable in the absence of the protein’s binding partner and solvent; hence providing a rationale for the importance of the gas-phase energy of the target protein relative to its unfolded state.

(3) As a result of these novel elements, our approach, PPI-HotspotID, could identify many true positives that were not detected by FTMap or SPOTONE (see Results and Figure 1). Previous methods generally predict residues that make multiple contacts across the proteinprotein interface as PPI-hot spots. In contrast, PPI-HotspotID can detect not only PPI-hot spots that make multiple contacts across the protein-protein interface, but also those lacking direct contact with the partner protein (see Discussion).

(4) Unlike most machine-learning methods which require feature customization, data preprocessing, and model optimization, our use of AutoGluon’s AutoTabular module

automates data preprocessing, model selection, hyperparameter optimization, and model evaluation. This automation reduces the need for manual intervention.

We have revised and added the following sentences on p. 9 in the Discussion section to highlight the novelty of our approach:

“Here, we have introduced two novel elements that have helped to identify PPI-hot spots using the unbound structure. First, we have constructed a dataset comprising 414 experimentally known PPI-hot spots and 504 nonhot spots, and carefully checked that PPI-hot spots have no mutations resulting in $\Delta\Delta G_{\text{bind}} < 0.5$ kcal/mol, whereas nonhot spots have no mutations resulting in $\Delta\Delta G_{\text{bind}} \geq 0.5$ kcal/mol or impact binding in immunoprecipitation or GST pull-down assays (see Methods). In contrast, SPOTONE53 employed nonhot spots defined as residues that upon alanine mutation resulted in $\Delta\Delta G_{\text{bind}} < 2.0$ kcal/mol. Notably, previous PPI-hot spot prediction methods did not employ PPIhot spots whose mutations have been curated to significantly impair/disrupt PPIs in UniProtKB (see Introduction). Second, we have introduced novel features derived from unbound protein structures such as the gas-phase energy of the target protein relative to its unfolded state.”

Strengths:

A new web server was developed for detecting protein-protein interaction hot spots.

Weaknesses:

The comparison to FTMap results is wrong. The method is not novel.

See reply to points 1 and 2 above.

Reviewer #2 (Public Review):

Summary:

The paper presents PPI-hotspot a method to predict PPI-hotspots. Overall, it could be useful but serious concerns about the validation and benchmarking of the methodology make it difficult to predict its reliability.

Strengths:

Develops an extended benchmark of hot-spots.

Weaknesses:

(1) Novelty seems to be just in the extended training set. Features and approaches have been used before.

The novelty of our approach extends beyond just the expanded training set, as summarized in our reply to Reviewer #1, point 2 above. To our knowledge, previous studies did not leverage the gas-phase energy of the target protein relative to its unfolded state for detecting PPI-hot spots from unbound structures. Previous studies did not automate the training and validation process. In contrast, we used AutoGluon’s AutoTabular module to automate the training of (i) individual “base” models, including LightGBM, CatBoost, XGBoost, random forests, extremely randomized trees, neural networks, and K-nearest neighbours, then (ii) multiple “stacker” models. The predictions of multiple “stacker” models were fed as inputs to additional higher layer stacker models in an iterative process called multi-layer stacking. The output layer used ensemble selection to aggregate the predictions of the stacker models. To improve stacking performance, AutoGluon used all the data for both training and validation through repeated k-fold bagging of all models at all layers of the stack, where k is determined

by best precision. This comprehensive approach, including repeated k-fold bagging of all models at all layers of the stack, sets our methodology apart from previous studies, including SPOTONE (see Methods).

(2) As far as I can tell the training and testing sets are the same. If I am correct, it is a fatal flaw.

The two most common techniques for testing and validating machine-learning methods are to split the dataset into either (1) a training set and a test set with a fixed ratio (e.g., 70% for training and 30% for testing) or (2) multiple subsets/folds; i.e., cross-validation. We did not employ a training set to train the model and a separate test set to evaluate its performance. Instead, we employed cross-validation, where the model was trained and evaluated multiple times. Each fold was used once as a test set and the remaining folds serve as the training set - this process was repeated for each fold. For each test set, we assessed the model's performance using the F1 score. We had listed the mean validation F1 score in Table 1 in the original manuscript. Cross-validation helps reduce the variability in performance estimates compared to a single training/test split. It also utilizes the entire dataset for training and testing, making efficient use of the limited data. We have clarified this on p. 14 in the revised version:

“AutoGluon was chosen for model training and validation due to its robustness and userfriendly interface, allowing for the simultaneous and automated exploration of various machine-learning approaches and their combinations. Instead of using a single training set to train the model and a separate test set to evaluate its performance, we employed cross-validation, as it utilizes the entire dataset for both training and testing, making efficient use of the limited data on PPI-hot spots and PPI-nonhot spots. AutoGluonTabular automatically chose a random partitioning of our dataset into multiple subsets/folds for training and validation. Notably, the training and validation data share insignificant homology, as the average pairwise sequence identity in our dataset is 26%. Each fold was used once as a test set, while the remaining folds served as the training set. For each test set, the model's performance was measured using the F1 score.”

(3) Comparisons should state that: SPOTONE is a sequence (only) based ML method that uses similar features but is trained on a smaller dataset. FTmap I think predicts binding sites, I don't understand how it can be compared with hot spots. Suggesting superiority by comparing with these methods is an overreach.

In the Introduction on page 3, we had already stated that:

“SPOTONE53 predicts PPI-hot spots from the protein sequence using residue-specific features such as atom type, amino acid (aa) properties, secondary structure propensity, and mass-associated values to train an ensemble of extremely randomized trees. The PPIhot spot prediction methods have mostly been trained, validated, and tested on data from the Alanine Scanning Energetics database (ASEdb)⁵⁵ and/or the Structural Kinetic and Energetic database of Mutant Protein Interactions (SKEMPI) 2.0 database.⁵⁶”

On p. 4, we have clarified how we used FTMAP to detect hot spots - see reply to Reviewer #1, point 1.

“We explored the possibility of detecting PPI-hot spots using the FTMap server in the PPI mode, which identifies hot spots on protein-protein interfaces from free protein structures.⁴⁵ These hot spots are identified by consensus sites – regions that bind multiple probe clusters.^{42,45,59} Such regions are deemed to be important for any interaction involving that region of the target, independent of partner protein.⁴² PPI-hot spots were identified as

residues in van der Waals (vdW) contact with probe ligands within the largest consensus site containing the most probe clusters.”

(4) Training in the same dataset as SPOTONE, and then comparing results in targets without structure could be valuable.

We think that the dataset used by SPOTONE is not as “clean” as ours since SPOTONE employed nonhot spots defined as aa residues that upon alanine mutation resulted in $\Delta\Delta G_{\text{bind}} < 2.0$ kcal/mol. In contrast, we define nonhot spots as residues whose mutations resulted in protein $\Delta\Delta G_{\text{bind}}$ changes < 0.5 kcal/mol. Moreover, we carefully checked that the nonhot spots have no mutations resulting in $\Delta\Delta G_{\text{bind}}$ changes ≥ 0.5 kcal/mol or impact binding in immunoprecipitation or GST pull-down assays (see Methods). We cannot compare results in targets without structure because we require the free protein structure to compute the perresidue free energy contributions.

(5) The paper presents as validation of the prediction and experimental validation of hotspots in human eEF2. Several predictions were made but only one was confirmed, what was the overall success rate of this exercise?

We did not test all predicted PPI-hot spots but only the PPI-hot spot with the highest probability of 0.67 (F794) and 7 other predicted PPI-hot spots that were > 12 Å from F794 as well as 4 predicted PPI-nonhot spots. Among the 13 predictions tested, F794 and the 4 predicted nonhot spots were confirmed to be correct.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Remove the comparison to FTMap, and find a more appropriate reference method, even if it requires installing programs rather than using the available web servers.

We have clarified comparison to FTMap in the revised ms - see our reply above.

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